

Linux Commands

Scope: Common Linux commands for system navigation, file management, permissions, users, processes, networking, searching, package management, logs, and basic security practice.

1. Navigation & Directory Management

Command	Description
pwd	Display the current working directory.
ls	List files and directories.
ls -la	List all files, including hidden files, with detailed information.
cd /path/to/directory	Change to a specific directory.
cd ..	Move to the parent directory.
cd ~	Move to the current user's home directory.
mkdir project	Create a directory named project.
mkdir -p project/logs	Create a directory and any missing parent directories.
rmdir project	Remove an empty directory.

2. File Management

Command	Description
touch file.txt	Create an empty file or update its timestamp.
cp file.txt backup.txt	Copy a file.
cp -r project project_backup	Copy a directory recursively.
mv old.txt new.txt	Move or rename a file.
rm file.txt	Remove a file.
rm -r project	Remove a directory and its contents recursively. Use carefully.
cat file.txt	Display file contents.
less file.txt	View a file page by page.
head file.txt	Show the beginning of a file.
tail file.txt	Show the end of a file.
tail -f /var/log/syslog	Follow a log file as new entries are added.

3. File Permissions & Ownership

Command	Description
<code>ls -l</code>	Display file permissions, ownership, size, and timestamps.
<code>chmod 644 file.txt</code>	Set owner read/write and group/others read permissions.
<code>chmod 755 script.sh</code>	Set owner read/write/execute and group/others read/execute.
<code>chmod 700 private_data</code>	Give the owner full access and no access to group/others.
<code>chmod +x script.sh</code>	Add execute permission.
<code>chmod -w file.txt</code>	Remove write permission.
<code>chown alice file.txt</code>	Change the owner of a file.
<code>chown alice:developers file.txt</code>	Change owner and group.
<code>chgrp developers file.txt</code>	Change the group owner.
<code>find /home -type f -perm -002</code>	Find files that are writable by others.

4. Users & Groups

Command	Description
<code>whoami</code>	Display the current username.
<code>id</code>	Show the current user's UID, GID, and group memberships.
<code>who</code>	Show users currently logged in.
<code>groups</code>	Show groups associated with the current user.
<code>sudo command</code>	Run a command with elevated privileges when permitted.
<code>passwd</code>	Change the current user's password.

5. Processes & System Information

Command	Description
<code>ps aux</code>	List running processes.
<code>top</code>	Display live process and resource usage information.
<code>htop</code>	Interactive process viewer, if installed.
<code>kill PID</code>	Send a termination signal to a process.
<code>uname -a</code>	Display kernel and system information.
<code>hostname</code>	Display the system hostname.
<code>uptime</code>	Show system uptime and load information.
<code>free -h</code>	Show memory usage in human-readable form.
<code>df -h</code>	Show filesystem disk usage.
<code>du -sh directory</code>	Show the size of a directory.

6. Networking

Command	Description
<code>ip addr</code>	Display network interfaces and IP addresses.
<code>ip link</code>	Display network interfaces and their state.
<code>ip route</code>	Display the routing table.
<code>ping 8.8.8.8</code>	Test basic network reachability.
<code>ss -tuln</code>	Show listening TCP/UDP sockets and ports.
<code>traceroute example.com</code>	Show the path packets take toward a destination.
<code>nslookup example.com</code>	Perform a DNS lookup.
<code>dig example.com</code>	Query DNS records in detail.
<code>curl -I https://example.com</code>	Request HTTP response headers.
<code>wget https://example.com/file</code>	Download a resource from a URL.

7. Searching & Text Processing

Command	Description
<code>grep 'error' logfile.txt</code>	Search a file for matching text.
<code>grep -i 'error' logfile.txt</code>	Search case-insensitively.
<code>grep -R 'password' /path</code>	Recursively search files for a string.
<code>find /home -name '*.txt'</code>	Find .txt files under /home.
<code>find / -type f -size +100M</code>	Find files larger than 100 MB.
<code>wc -l file.txt</code>	Count lines in a file.
<code>sort file.txt</code>	Sort lines.

Command	Description
<code>uniq file.txt</code>	Remove adjacent duplicate lines.
<code>cut -d: -f1 /etc/passwd</code>	Extract the first colon-separated field.

8. Archives & Compression

Command	Description
<code>tar -cvf archive.tar directory/</code>	Create a tar archive.
<code>tar -xvf archive.tar</code>	Extract a tar archive.
<code>tar -czvf archive.tar.gz directory/</code>	Create a gzip-compressed tar archive.
<code>tar -xzvf archive.tar.gz</code>	Extract a gzip-compressed tar archive.
<code>gzip file.txt</code>	Compress a file using gzip.
<code>gunzip file.txt.gz</code>	Decompress a gzip file.

9. Package Management — Debian/Kali

Command	Description
<code>sudo apt update</code>	Refresh available package information.
<code>sudo apt upgrade</code>	Upgrade installed packages.
<code>sudo apt install package</code>	Install a package.
<code>sudo apt remove package</code>	Remove a package.
<code>apt search keyword</code>	Search available packages.
<code>apt show package</code>	Display package information.

10. Services & Logs

Command	Description
<code>systemctl status service</code>	Check the status of a systemd service.
<code>sudo systemctl start service</code>	Start a service.
<code>sudo systemctl stop service</code>	Stop a service.
<code>sudo systemctl restart service</code>	Restart a service.
<code>journalctl</code>	View systemd journal logs.
<code>journalctl -u service</code>	View logs for a specific service.
<code>dmesg</code>	Display kernel ring-buffer messages.

11. Security & Troubleshooting

Command	Description
<code>history</code>	Display previously executed shell commands.
<code>man command</code>	Open the manual page for a command.
<code>command --help</code>	Show common usage information for a command.
<code>sudo lsof -i</code>	Show processes using network connections, when lsof is installed.
<code>file filename</code>	Identify the type of a file.
<code>sha256sum file.iso</code>	Calculate a SHA-256 checksum.
<code>chmod 600 confidential.txt</code>	Restrict a file to owner read/write access.
<code>chmod 700 backup.sh</code>	Restrict a script to owner read/write/execute access.

Linux Permissions Reference

Common permission values and their practical meaning.

Command	Description
r	Read — numeric value 4.
w	Write — numeric value 2.
x	Execute — numeric value 1.
rwX	Read, write, and execute — numeric value 7.
rw-	Read and write — numeric value 6.
r-x	Read and execute — numeric value 5.
r--	Read only — numeric value 4.
--x	Execute only — numeric value 1.
---	No permissions — numeric value 0.

Common Permission Settings

Command	Description
777	Full access for everyone; generally avoid unless there is a specific requirement.
755	Owner has full access; group and others have read/execute.
744	Owner has full access; group and others have read.
700	Owner has full access; group and others have no access.
644	Owner has read/write; group and others have read.
600	Owner has read/write; group and others have no access.

Permission Example

```
-rw-r--r-- 1 user user 1200 Jun 23 10:00 file.txt
```

The first character indicates the file type. The next three characters represent the owner's permissions, followed by the group's permissions and then the permissions for others.

Practical Safety Notes

- Use `man` command or `command --help` to understand unfamiliar options.
- Use `sudo` carefully because elevated privileges can affect the entire system.
- Be particularly careful with recursive deletion such as `rm -r`.
- Choose file permissions based on the minimum access required.